

The Shape of Exploitation and Vulnerability

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1 Introduction

For any computational substrate which implements a simple interface, you can produce an adversarial game of construction and exploitation.

2 Interface

The substrate of choice must be computational, in that you must be able to take a term A and naturally derive its evaluation to a term B , it must allow terms which take input and produce output, and it must be able to encode, whether artificially or intrinsically, side effectful computations. Additionally, the substrate must be able to encode an identity term.

3 Shape

A player in defense of their position must construct a term F which encodes a series of constraints C , and an optional filter term S such that $F \circ S$ encodes all constraints in C . Some constraints are designed to not be exhaustively provable, but for the constraints which are, proofs that $F \circ S$ fulfil those constraints are required. A default identity term is chosen for S .

4 Constraints

While this is malleable based on the instantiated substrate, a sensible set of constraint templates follow:

$$\exists x \text{ s.t. } (F \circ S)(x) = A$$

$$\neg \exists x \text{ s.t. } (F \circ S)(x) = A$$

$$\forall x, (F \circ S)(x) \text{ produces side effect } A$$

$$\forall x, (F \circ S)(x) \text{ does not produce side effect } A$$